# Dirichlet's Theorem on Primes in Arithmetic Progressions

UGP Presentation (MTH497)

Saptarshi Mandal

Indian Institute of Technology, Kanpur

November 7, 2026

**Mentor**: Dr. Somnath Jha

# Primes in arithmetic progressions

**Euclid:** there are infinitely many primes.

# Primes in arithmetic progressions

**Euclid:** there are infinitely many primes.

**Finer question.** Take integers $a, m \geq 1$ and look at

$$a,\ a+m,\ a+2m,\ a+3m,\ \ldots$$

Does this list contain primes? Infinitely many?

# Primes in arithmetic progressions

**Euclid:** there are infinitely many primes.

**Finer question.** Take integers $a, m \geq 1$ and look at

$$a,\ a+m,\ a+2m,\ a+3m,\ \ldots$$

Does this list contain primes? Infinitely many?

**Examples.**

- $a = 3,\ m = 4$:
  $3,\ 7,\ 11,\ 19,\ 23,\ 31, \ldots$
- $a = 1,\ m = 4$:
  $5,\ 13,\ 17,\ 29,\ 37, \ldots$

**A necessary condition.** If $\gcd(a, m) = d > 1$, every term is divisible by $d$, so there is at most one prime.

# Dirichlet's theorem

**Theorem (Dirichlet)**

*If* $\gcd(a, m) = 1$, *there are infinitely many primes* $p \equiv a \pmod{m}$.

# Dirichlet's theorem

**Theorem (Dirichlet)**

*If* $\gcd(a, m) = 1$, *there are infinitely many primes* $p \equiv a \pmod{m}$.

In fact the primes are equidistributed among the $\varphi(m)$ admissible classes.

# Dirichlet's theorem

**Theorem (Dirichlet)**

*If* $\gcd(a, m) = 1$, *there are infinitely many primes* $p \equiv a \pmod{m}$.

In fact the primes are equidistributed among the $\varphi(m)$ admissible classes.

**Density**

A set $A$ of primes has *density* $k$ if $\sum_{p \in A} p^{-s} \Big/ \log \frac{1}{s-1} \to k$ as $s \to 1^+$.

**Theorem (Dirichlet, density form)**

$P_a = \{p \text{ prime} : p \equiv a \pmod{m}\}$ *has density* $\dfrac{1}{\varphi(m)}$.

# Euler's idea: primes via the zeta function

For $\mathrm{Re}(s) > 1$, unique factorisation gives the Euler product

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_{p} \frac{1}{1 - p^{-s}}.$$

# Euler's idea: primes via the zeta function

For $\mathrm{Re}(s) > 1$, unique factorisation gives the Euler product

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_{p} \frac{1}{1 - p^{-s}}.$$

- As $s \to 1^+$: $\zeta(s) \sim \dfrac{1}{s-1} \to \infty$.

# Euler's idea: primes via the zeta function

For $\mathrm{Re}(s) > 1$, unique factorisation gives the Euler product

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_{p} \frac{1}{1 - p^{-s}}.$$

- As $s \to 1^+$: $\zeta(s) \sim \dfrac{1}{s-1} \to \infty$.
- Take logarithms: $\log \zeta(s) = \sum_p p^{-s} + (\text{bounded})$.

# Euler's idea: primes via the zeta function

For $\mathrm{Re}(s) > 1$, unique factorisation gives the Euler product

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_{p} \frac{1}{1 - p^{-s}}.$$

- As $s \to 1^+$: $\zeta(s) \sim \dfrac{1}{s-1} \to \infty$.
- Take logarithms: $\log \zeta(s) = \sum_p p^{-s} + (\text{bounded})$.
- Hence

$$\sum_{p} p^{-s} \;\sim\; \log \frac{1}{s-1} \qquad (s \to 1^+),$$

so the set of all primes has density 1.

# Euler's idea: primes via the zeta function

For $\mathrm{Re}(s) > 1$, unique factorisation gives the Euler product

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_{p} \frac{1}{1 - p^{-s}}.$$

- As $s \to 1^+$: $\zeta(s) \sim \dfrac{1}{s-1} \to \infty$.
- Take logarithms: $\log \zeta(s) = \sum_p p^{-s} + (\text{bounded})$.
- Hence

$$\sum_{p} p^{-s} \;\sim\; \log \frac{1}{s-1} \qquad (s \to 1^+),$$

so the set of all primes has density 1.

**Idea**

Isolate primes $\equiv a \pmod{m}$ inside this sum, using **characters**.

# Step 1: characters of a finite abelian group

**Definition**

Let $G$ be a finite abelian group. A *character* is a homomorphism $\chi : G \to \mathbb{C}^\times$.
The characters form a group $\widehat{G}$ under pointwise multiplication.

# Step 1: characters of a finite abelian group

**Definition**

Let $G$ be a finite abelian group. A *character* is a homomorphism $\chi : G \to \mathbb{C}^\times$.
The characters form a group $\widehat{G}$ under pointwise multiplication.

**Proposition (Orthogonality)**

*Let $n = |G|$. Then*

$$\sum_{x \in G} \chi(x) = \begin{cases} n & \chi = 1 \\ 0 & \chi \neq 1 \end{cases} \qquad \sum_{\chi \in \widehat{G}} \chi(x) = \begin{cases} n & x = 1 \\ 0 & x \neq 1. \end{cases}$$

# Step 1: characters of a finite abelian group

**Definition**

Let $G$ be a finite abelian group. A *character* is a homomorphism $\chi : G \to \mathbb{C}^\times$.
The characters form a group $\widehat{G}$ under pointwise multiplication.

**Proposition (Orthogonality)**

*Let $n = |G|$. Then*

$$\sum_{x \in G} \chi(x) = \begin{cases} n & \chi = 1 \\ 0 & \chi \neq 1 \end{cases} \qquad \sum_{\chi \in \widehat{G}} \chi(x) = \begin{cases} n & x = 1 \\ 0 & x \neq 1. \end{cases}$$

So for $a, b \in G$, we can write $\mathbf{1_{a=b}} = \frac{1}{|G|} \sum_{\chi \in \hat{G}} \overline{\chi(a)}\chi(b) = \begin{cases} 1 & a = b \\ 0 & a \neq b \end{cases}$.

# Dirichlet characters

Take $G = U_m = (\mathbb{Z}/m\mathbb{Z})^\times$, of order $\varphi(m)$. A character $\chi \in \widehat{U_m}$ is extended to $\mathbb{Z}$ by $\chi(n) = 0$ if $\gcd(n, m) > 1$. Then $\chi$ is completely multiplicative and $m$-periodic.

# Dirichlet characters

Take $G = U_m = (\mathbb{Z}/m\mathbb{Z})^\times$, of order $\varphi(m)$. A character $\chi \in \widehat{U_m}$ is extended to $\mathbb{Z}$ by $\chi(n) = 0$ if $\gcd(n, m) > 1$. Then $\chi$ is completely multiplicative and $m$-periodic.

**Example**

$m = 4$: the non-trivial character is

$$\chi_4(n) = \begin{cases} 1 & n \equiv 1 \pmod 4 \\ -1 & n \equiv 3 \pmod 4 \\ 0 & n \text{ even.} \end{cases}$$

# Dirichlet characters

Take $G = U_m = (\mathbb{Z}/m\mathbb{Z})^\times$, of order $\varphi(m)$. A character $\chi \in \widehat{U_m}$ is extended to $\mathbb{Z}$ by $\chi(n) = 0$ if $\gcd(n, m) > 1$. Then $\chi$ is completely multiplicative and $m$-periodic.

**Example**

$m = 4$: the non-trivial character is

$$\chi_4(n) = \begin{cases} 1 & n \equiv 1 \pmod 4 \\ -1 & n \equiv 3 \pmod 4 \\ 0 & n \text{ even.} \end{cases}$$

Applying the "detector" $(\mathbf{1}_{a=b})$ to $b = p$ gives, for $\gcd(a, m) = 1$,

$$\mathbf{1}_{p\equiv a} \;=\; \frac{1}{\varphi(m)} \sum_{\chi} \overline{\chi(a)}\, \chi(p).$$

## Step 2: reduction to the characters

Define $f_\chi(s) = \sum_{p \nmid m} \chi(p)\, p^{-s}$. Summing the "detector" over primes:

$$\sum_{p \equiv a\,(m)} p^{-s} \;=\; \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{U_m}} \overline{\chi(a)}\, f_\chi(s).$$

## Step 2: reduction to the characters

Define $f_\chi(s) = \sum_{p\nmid m} \chi(p)\, p^{-s}$. Summing the "detector" over primes:

$$\sum_{p\equiv a\,(m)} p^{-s} \;=\; \frac{1}{\varphi(m)} \sum_{\chi\in\widehat{U_m}} \overline{\chi(a)}\, f_\chi(s).$$

- $\chi = 1$: $f_1(s) \sim \log \frac{1}{s-1}$ (Euler's computation).

## Step 2: reduction to the characters

Define $f_\chi(s) = \sum_{p \nmid m} \chi(p)\, p^{-s}$. Summing the "detector" over primes:

$$\sum_{p \equiv a\,(m)} p^{-s} \;=\; \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{U_m}} \overline{\chi(a)}\, f_\chi(s).$$

- $\chi = 1$: $f_1(s) \sim \log \frac{1}{s-1}$ (Euler's computation).
- $\chi \neq 1$: we want $f_\chi(s)$ to stay bounded as $s \to 1^+$.

# Step 2: reduction to the characters

Define $f_\chi(s) = \sum_{p \nmid m} \chi(p)\, p^{-s}$. Summing the "detector" over primes:

$$\sum_{p \equiv a\,(m)} p^{-s} \;=\; \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{U_m}} \overline{\chi(a)}\, f_\chi(s).$$

- $\chi = 1$: $f_1(s) \sim \log \frac{1}{s-1}$ (Euler's computation).
- $\chi \neq 1$: we want $f_\chi(s)$ to stay bounded as $s \to 1^+$.

**If this holds...**

Dividing by $\log \frac{1}{s-1}$ and letting $s \to 1^+$, only $\chi = 1$ survives:

$$\text{density of } P_a = \frac{1}{\varphi(m)}.$$

## Step 3: $L$-functions

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_{p\nmid m} \frac{1}{1-\chi(p)p^{-s}} \qquad (\mathrm{Re}\, s > 1).$$

## Step 3: $L$-functions

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_{p\nmid m} \frac{1}{1-\chi(p)p^{-s}} \qquad (\operatorname{Re} s > 1).$$

Taking logarithms: $\log L(s,\chi) = f_\chi(s) + (\text{bounded})$. So

$$f_\chi \text{ bounded as } s \to 1^+ \iff \log L(s,\chi) \text{ bounded.}$$

## Step 3: $L$-functions

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_{p\nmid m} \frac{1}{1-\chi(p)p^{-s}} \qquad (\operatorname{Re} s > 1).$$

Taking logarithms: $\log L(s,\chi) = f_\chi(s) + (\text{bounded})$. So

$$f_\chi \text{ bounded as } s \to 1^+ \iff \log L(s,\chi) \text{ bounded.}$$

- For $\chi \neq 1$, the partial sums of $\chi(n)$ stay bounded (orthogonality), so $L(s,\chi)$ converges and is holomorphic for $\operatorname{Re}(s) > 0$.

## Step 3: $L$-functions

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_{p\nmid m} \frac{1}{1-\chi(p)p^{-s}} \qquad (\mathrm{Re}\, s > 1).$$

Taking logarithms: $\log L(s,\chi) = f_\chi(s) + (\text{bounded})$. So

$$f_\chi \text{ bounded as } s \to 1^+ \iff \log L(s,\chi) \text{ bounded.}$$

- For $\chi \neq 1$, the partial sums of $\chi(n)$ stay bounded (orthogonality), so $L(s,\chi)$ converges and is holomorphic for $\mathrm{Re}(s) > 0$.
- Hence $L(s,\chi) \to L(1,\chi)$, a finite number. The log is bounded exactly when

$$L(1,\chi) \neq 0.$$

# Step 3: $L$-functions

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} = \prod_{p\nmid m} \frac{1}{1-\chi(p)p^{-s}} \qquad (\mathrm{Re}\, s > 1).$$

Taking logarithms: $\log L(s,\chi) = f_\chi(s) + (\text{bounded})$. So

$$f_\chi \text{ bounded as } s \to 1^+ \iff \log L(s,\chi) \text{ bounded.}$$

- For $\chi \neq 1$, the partial sums of $\chi(n)$ stay bounded (orthogonality), so $L(s,\chi)$ converges and is holomorphic for $\mathrm{Re}(s) > 0$.
- Hence $L(s,\chi) \to L(1,\chi)$, a finite number. The log is bounded exactly when

$$L(1,\chi) \neq 0.$$

*Example:* $L(1,\chi_4) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4} \neq 0.$

# Step 4: the key theorem

**Theorem**

*For every non-trivial Dirichlet character $\chi$, $L(1, \chi) \neq 0$.*

# Step 4: the key theorem

**Theorem**

*For every non-trivial Dirichlet character $\chi$, $L(1,\chi) \neq 0$.*

**Strategy (Dirichlet–Landau).** Suppose $L(1,\chi_0) = 0$ for some $\chi_0 \neq 1$ and consider the product of *all* $L$-functions mod $m$:

$$\zeta_m(s) = \prod_{\chi \in \widehat{U_m}} L(s,\chi).$$

# Step 4: the key theorem

**Theorem**

*For every non-trivial Dirichlet character $\chi$, $L(1,\chi) \neq 0$.*

**Strategy (Dirichlet–Landau).** Suppose $L(1,\chi_0) = 0$ for some $\chi_0 \neq 1$ and consider the product of *all* $L$-functions mod $m$:

$$\zeta_m(s) = \prod_{\chi \in \widehat{U_m}} L(s,\chi).$$

- $L(s,1)$ has a simple pole at $s = 1$ (it is $\zeta(s)$ up to finitely many Euler factors).

# Step 4: the key theorem

**Theorem**

*For every non-trivial Dirichlet character $\chi$, $L(1,\chi) \neq 0$.*

**Strategy (Dirichlet–Landau).** Suppose $L(1,\chi_0) = 0$ for some $\chi_0 \neq 1$ and consider the product of *all* $L$-functions mod $m$:

$$\zeta_m(s) = \prod_{\chi \in \widehat{U_m}} L(s,\chi).$$

- $L(s,1)$ has a simple pole at $s = 1$ (it is $\zeta(s)$ up to finitely many Euler factors).
- The zero of $L(s,\chi_0)$ cancels this pole. Every other factor is holomorphic on $\operatorname{Re} s > 0$.

# Step 4: the key theorem

**Theorem**

*For every non-trivial Dirichlet character $\chi$, $L(1,\chi) \neq 0$.*

**Strategy (Dirichlet–Landau).** Suppose $L(1,\chi_0) = 0$ for some $\chi_0 \neq 1$ and consider the product of *all* $L$-functions mod $m$:

$$\zeta_m(s) = \prod_{\chi \in \widehat{U_m}} L(s,\chi).$$

- $L(s,1)$ has a simple pole at $s = 1$ (it is $\zeta(s)$ up to finitely many Euler factors).
- The zero of $L(s,\chi_0)$ cancels this pole. Every other factor is holomorphic on $\operatorname{Re} s > 0$.
- So $\zeta_m(s)$ would be holomorphic on the whole half-plane $\operatorname{Re}(s) > 0$.

# Proof of $L(1,\chi) \neq 0$: the contradiction

**Fact.** $\zeta_m(s) = \sum c_n n^{-s}$ has non-negative coefficients.
Landau's theorem applied to $\zeta_m(s) = \sum c_n n^{-s}$ (holomorphic on $\operatorname{Re} s > 0$) shows:

the series converges for *all* real $s > 0$.

# Proof of $L(1,\chi) \neq 0$: the contradiction

**Fact.** $\zeta_m(s) = \sum c_n n^{-s}$ has non-negative coefficients.
Landau's theorem applied to $\zeta_m(s) = \sum c_n n^{-s}$ (holomorphic on $\operatorname{Re} s > 0$) shows:

the series converges for *all* real $s > 0$.

But for $\gcd(n, m) = 1$, expanding the Euler product gives $c_{n^{\varphi(m)}} \geq 1$. At $s = \frac{1}{\varphi(m)}$:

$$\sum_N \frac{c_N}{N^s} \geq \sum_{\gcd(n,m)=1} \frac{1}{n} = \infty.$$

# Proof of $L(1,\chi) \neq 0$: the contradiction

**Fact.** $\zeta_m(s) = \sum c_n n^{-s}$ has non-negative coefficients.
Landau's theorem applied to $\zeta_m(s) = \sum c_n n^{-s}$ (holomorphic on $\operatorname{Re} s > 0$) shows:

the series converges for *all* real $s > 0$.

But for $\gcd(n, m) = 1$, expanding the Euler product gives $c_{n^{\varphi(m)}} \geq 1$. At $s = \frac{1}{\varphi(m)}$:

$$\sum_N \frac{c_N}{N^s} \geq \sum_{\gcd(n,m)=1} \frac{1}{n} = \infty.$$

**Contradiction**

The series diverges at $s = 1/\varphi(m) > 0$. Hence $L(1,\chi) \neq 0$ for every $\chi \neq 1$.

# Putting it all together

1. $L(1,\chi) \neq 0 \implies \log L(s,\chi)$ is bounded as $s \to 1^+$.

# Putting it all together

1. $L(1,\chi) \neq 0 \implies \log L(s,\chi)$ is bounded as $s \to 1^+$.
2. $\implies f_\chi(s) = \sum_{p \nmid m} \chi(p) p^{-s}$ is bounded for $\chi \neq 1$.

# Putting it all together

1. $L(1,\chi) \neq 0 \implies \log L(s,\chi)$ is bounded as $s \to 1^+$.
2. $\implies f_\chi(s) = \sum_{p \nmid m} \chi(p) p^{-s}$ is bounded for $\chi \neq 1$.
3. $\implies$ in the identity

$$\sum_{p \equiv a} p^{-s} = \frac{1}{\varphi(m)} \sum_{\chi} \overline{\chi(a)} f_\chi(s)$$

only $f_1 \sim \log \frac{1}{s-1}$ contributes.

# Putting it all together

1. $L(1,\chi) \neq 0 \implies \log L(s,\chi)$ is bounded as $s \to 1^+$.
2. $\implies f_\chi(s) = \sum_{p \nmid m} \chi(p) p^{-s}$ is bounded for $\chi \neq 1$.
3. $\implies$ in the identity

$$\sum_{p \equiv a} p^{-s} = \frac{1}{\varphi(m)} \sum_{\chi} \overline{\chi(a)} f_\chi(s)$$

   only $f_1 \sim \log \frac{1}{s-1}$ contributes.

4. $\implies P_a$ has density $\dfrac{1}{\varphi(m)}$. In particular it is infinite. ■

Thank You!

# Thank You!

questions?